

IKENNA UDEANI

Data Scientist | Machine Learning · Anomaly Detection · Spatiotemporal & Graph Modelling

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PROFILE

Data scientist who takes machine learning from messy, real-world data to explainable models in production. Built **unsupervised anomaly detection** that flagged unusual maternal and neonatal mortality figures for data quality assessment at the Clinton Health Access Initiative (Nigeria), **cutting manual data reconciliation by 90%**, and **Provenance**, an explainable AI system that separates genuine pollution events from faulty sensor readings with gradient boosting, graph neural networks, SHAP and conformal prediction. MSc Data Science candidate researching spatiotemporal graph neural networks for outbreak detection. Explains assumptions, uncertainty and limitations clearly to non-technical stakeholders.

TECHNICAL SKILLS

Machine Learning: supervised and unsupervised learning, classification, regression, gradient boosting (LightGBM), anomaly detection (Isolation Forest, Local Outlier Factor, One-Class SVM), clustering (K-Means), time series forecasting, deep learning, graph neural networks, feature engineering, statistical analysis, exploratory data analysis

Evaluation & MLOps: time-blocked cross-validation, evaluation metrics, explainable AI (SHAP), uncertainty quantification (conformal prediction), model cards, drift monitoring, reproducible pipelines, Docker, CI, pytest, Git/GitHub

Programming: Python (pandas, NumPy, scikit-learn, PyTorch, PyTorch Geometric, TensorFlow/Keras, Streamlit, Plotly), SQL, R, FastAPI, LLM APIs

Data & Cloud: ETL/ELT pipelines, Apache Spark/PySpark, PostgreSQL/PostGIS, TimescaleDB, SQL Server, Neo4j, Azure (Synapse, Databricks), AWS (EMR), Snowflake, Power BI (DAX), Tableau

Domain: public health surveillance, longitudinal patient-level data, DHIS2, laboratory information systems (LIMS), geospatial and sensor data

PROFESSIONAL EXPERIENCE

Data Analyst, Maternal & Newborn Health (CBHMIS)

Dec 2024 – Oct 2025

Clinton Health Access Initiative (CHAI), Abuja, Nigeria

- Built **unsupervised anomaly detection** in Python (**Isolation Forest**, cross-checked with **Local Outlier Factor** and **One-Class SVM**) to flag wards and LGAs with statistically unusual maternal and neonatal mortality rates in Kaduna and Gombe, as part of **data quality assessment (DQA)**.
- Engineered rate features (deaths per 100,000 and per 1,000 live births) and **K-Means** clusters of wards, surfacing **data quality** errors such as deaths reported against zero live births, and **cut manual data reconciliation by 90%**.
- Delivered **Streamlit** dashboards ranking flagged areas for field verification, owned CBHMIS **Power BI** reporting for CHAI leadership and **Gates Foundation** donor teams, and helped the Federal Ministry of Health activate **DHIS2** Community HMIS nationwide.

Consultant, Data Management & Health Informatics

Oct 2023 – Nov 2024

Clinton Health Access Initiative (CHAI), Abuja, Nigeria

- Unified data pipelines and KPI definitions across **4 national diagnostic programmes (TB, HIV, HPV, EID)**, turning fragmented source systems into analysis-ready datasets; trained lab scientists and officials nationwide in data analysis.
- Administered Nigeria's **National LIMS** and led the HPV module upgrade from requirements (SRS) through testing to **national rollout**.

Data Engineer, Health Informatics

Feb 2023 – Sep 2023

Clinton Health Access Initiative (CHAI), Abuja, Nigeria

- Automated **Python, Spark and SQL ETL pipelines** for TB Diagnostic Network Optimization (specimen referral routes, testing capacity, turnaround times) and shipped **Power BI dashboards** for 3 national projects.

Database Developer

Jan 2021 – Dec 2021

Bonitas ICT, Port Harcourt, Nigeria

- Designed a **3NF relational schema** for coiled-tubing asset data, tuned indexes and queries, and documented to ISO 9001 (Agile, Jira).

SELECTED PROJECTS

Provenance: explainable AI trust layer for environmental sensor networks

2026

DEIK.AI Challenge 2026 · Python, LightGBM, PyTorch Geometric, SHAP, FastAPI, PostGIS, Docker · provenancel2.com

- Audited **149,683 readings** from 16 city sensor stations with 16 statistical detectors, exposing a **29% defect rate** behind 100% completeness.
- Built **LightGBM** deweathering and fault-classification models with **time-blocked cross-validation**, with **SHAP** explanations served via **FastAPI**.
- Built **HST-GAT**, a spatiotemporal **graph attention network** with **conformal prediction intervals** (90% coverage) and drift monitoring.

MSc thesis: spatiotemporal GNNs for early outbreak detection

2025 – 2027

University of Debrecen · PyTorch, GNNs, Google COVID-19 Open Data (U.S.)

- Training **GCN, GAT and temporal GNNs** on dynamic mobility graphs to predict regional infection risk, with **Isolation Forest** anomaly detection.

LinkWork (DEIK.AI 2026): hiring platform with a public match ledger and **LLM-assisted interview scoring** (Claude API); React, Express, SQLite.

EDUCATION

MSc, Data Science (EQF 7)

Sep 2025 – 2027 (exp.)

University of Debrecen, Hungary · Erasmus semester at the University of Toulouse, France (2026)

BEng, Petroleum and Gas Engineering (EQF 6), Second Class Upper · University of Port Harcourt, Nigeria

2015 – 2021

CERTIFICATIONS & LANGUAGES

Certifications: Power BI Data Analyst (PL-300) · Azure AI & Data Fundamentals · AWS Cloud Practitioner · Stanford SQL · McKinsey Forward

Languages: English (native) · French (B1, improving daily) · Hungarian (A1)